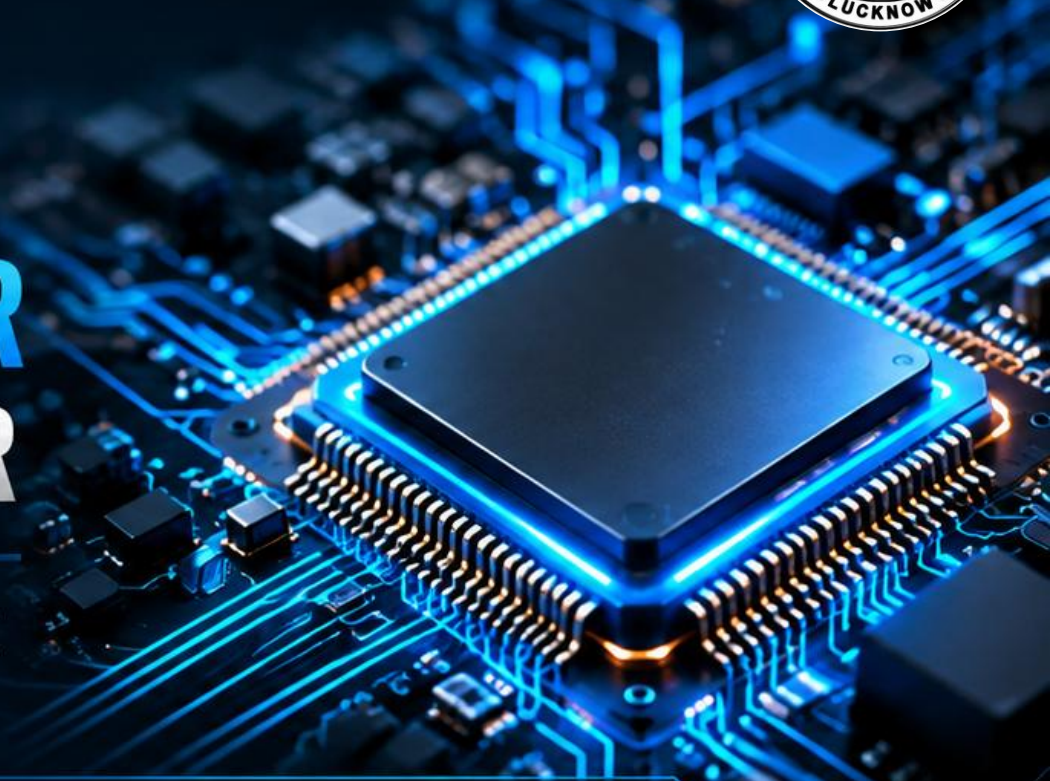
INSTITUTE OF ENGINEERING & TECHNOLOGY
LUCKNOW

PROJECT TITLE

CACHE MEMORY SIMULATOR AND PERFORMANCE ANALYZER



SUBJECT: COMPUTER ORGANIZATION AND ARCHITECTURE

SUBMITTED BY:

OM SHIVAM ASHEESH

MCA 1st SEMESTER

ROLL NO.: 2600520140049

SUBMITTED TO:

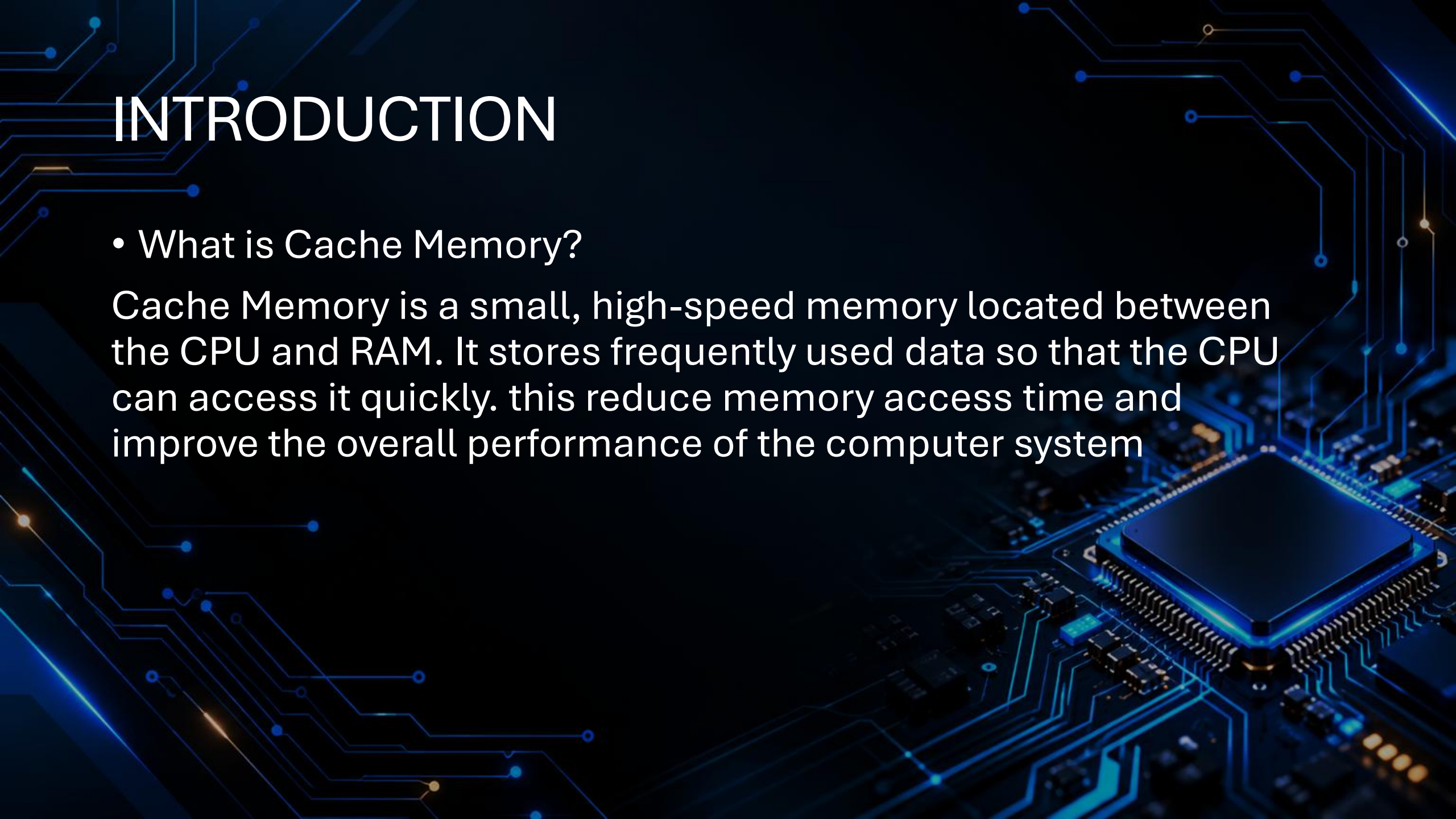
MRS. SHWETA TIWARI

DEPARTMENT OF COMPUTER SCIENCE

INTRODUCTION

- What is Cache Memory?

Cache Memory is a small, high-speed memory located between the CPU and RAM. It stores frequently used data so that the CPU can access it quickly. this reduce memory access time and improve the overall performance of the computer system



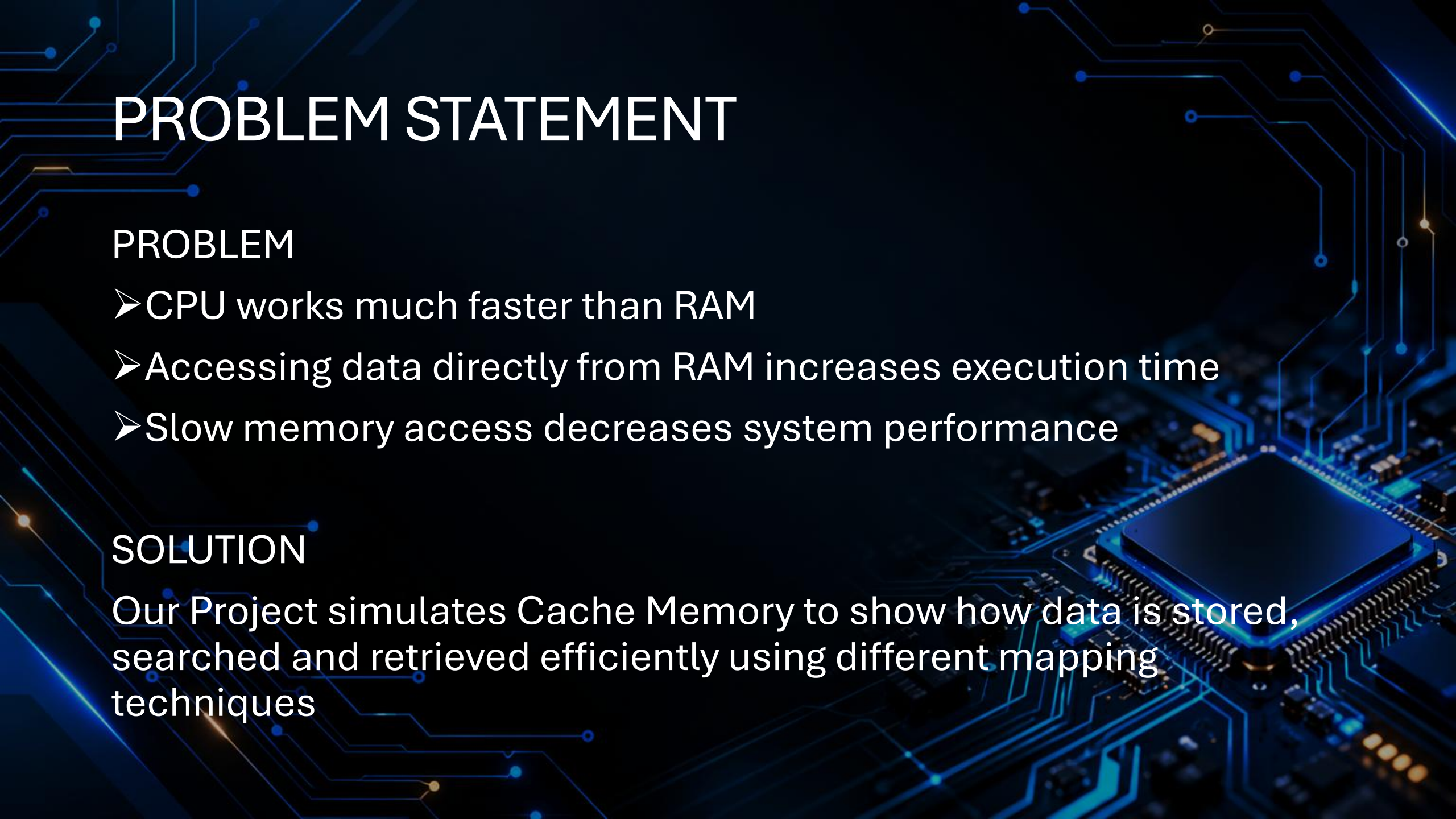
PROBLEM STATEMENT

PROBLEM

- CPU works much faster than RAM
- Accessing data directly from RAM increases execution time
- Slow memory access decreases system performance

SOLUTION

Our Project simulates Cache Memory to show how data is stored, searched and retrieved efficiently using different mapping techniques



OBJECTIVES

- To understand the concept of Cache Memory.
- To simulate cache operations
- To demonstrate Direct, Fully Associative, and set Associative Mapping
- To identify cache hits and cache misses
- To calculate hit ratio and miss ratio



TECHNOLOGIES USED

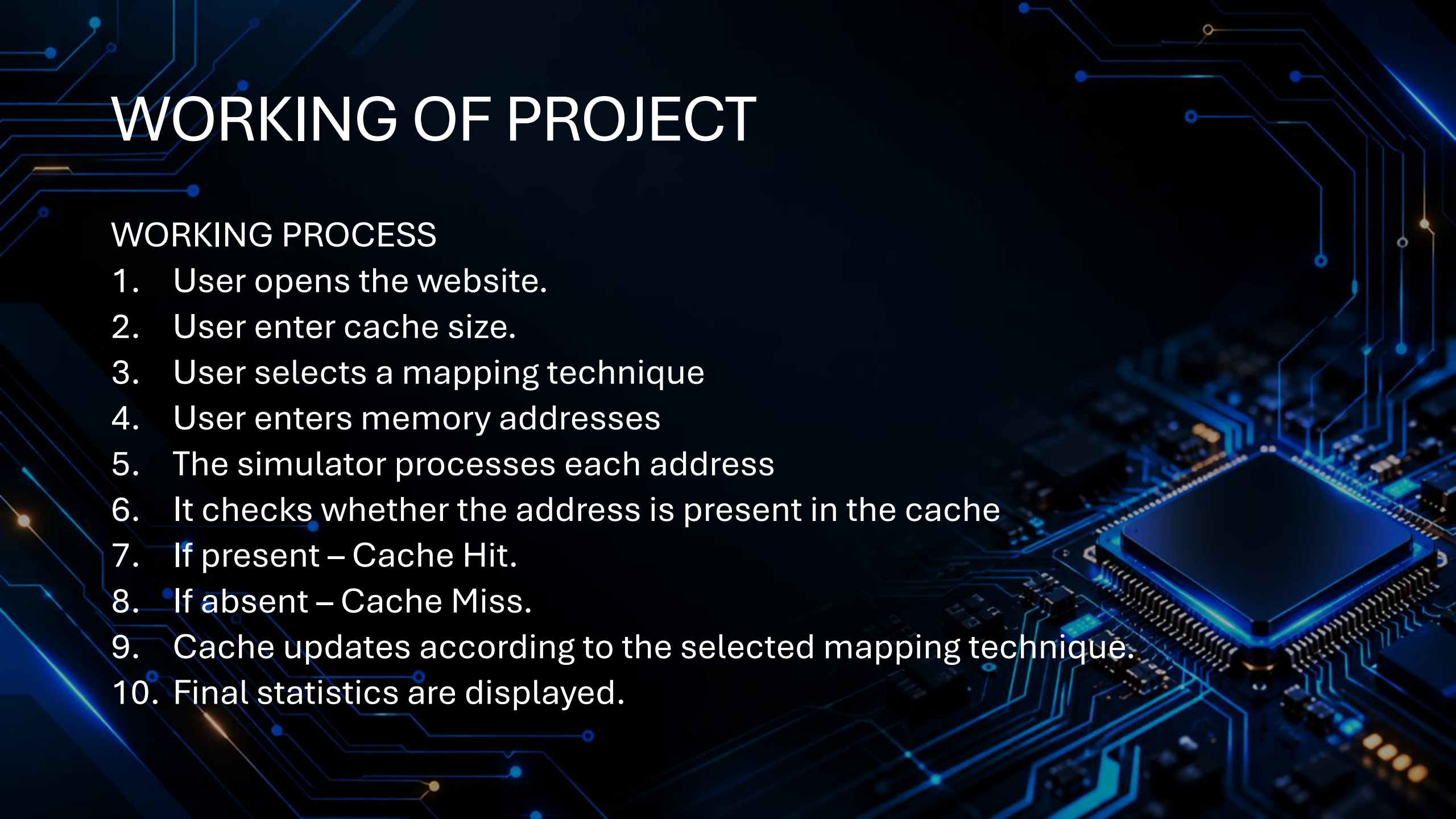
TECHNOLOGY	PURPOSE
• HTML	• STRUCTURE OF WEBSITE
• CSS	• STYLING AND DESIGN
• JAVA SCRIPT	• SIMULATION LOGIC
• VS CODE	• CODE EDITOR



WORKING OF PROJECT

WORKING PROCESS

1. User opens the website.
2. User enter cache size.
3. User selects a mapping technique
4. User enters memory addresses
5. The simulator processes each address
6. It checks whether the address is present in the cache
7. If present – Cache Hit.
8. If absent – Cache Miss.
9. Cache updates according to the selected mapping technique.
10. Final statistics are displayed.



SIMULATION RESULT

Output

The simulator displays:

- Cache Hit
- Cache Miss
- Cache Content
- Total Hits
- Total Misses
- Hit Ratio
- Miss Ratio



Mapping Techniques

- Direct Mapping

Each memory block maps to one fixed cache block.

- Fully Associative Mapping

A memory blocks can be placed in any cache block.

- Set Associative Mapping

A memory block maps to a specific set but can occupy any block within that set

CONCLUSION

This project demonstrates the working of cache memory through an interactive web application it helps user understand different cache mapping techniques and analyze cache performance using Hit Ratio and Miss Ratio.

